Context & Problem

As Supervisor, Onsite IT & Automation Systems, I inherited a manual data-prep process: the Americas region relied on manually populated Excel data for quarterly chargeback processing. These files:

- arrived in inconsistent formats
- contained variable column names
- required manual cleanup
- were stored across multiple SharePoint folders
- demanded hours of repetitive preparation every quarter

This created:

- high operational overhead
- inconsistent data quality
- risk of human error
- delays in financial reconciliation
- no reusable ingestion pattern

The process was not scalable and could not support automation.

Objective

Design a **reusable, governed ingestion framework** inside Power Query that:

- automatically pulls completed chargeback data from monthly or quarterly report files.
- normalizes schema differences
- applies consistent transformations
- produces clean, analysis-ready data
- minimizes manual preparation
- supports future automation and reporting

What I Built

a. Multi-File SharePoint Ingestion Pipeline

- Connected Power Query directly to credentialed SharePoint Online document libraries
- Implemented pattern-based filtering to ingest only relevant quarterly files
- Enabled dynamic ingestion without modifying queries each quarter

b. Schema Normalization Layer

- Standardized column names across the internal source files
- Applied type enforcement to prevent downstream errors
- Introduced a reusable “column harmonization” function

c. Transformation & Cleanup Layer

- Removed null rows, duplicate entries, and file-specific inconsistencies
- Applied business rules consistently across all files
- Added calculated fields required for chargeback logic

d. Reconciliation-Ready Output

- Produced a clean, unified dataset
- Ensured compatibility with downstream Excel models
- Structured the output for year-end validation against the vendor's report
- Enabled repeatable quarterly ingestion with minimal manual prep

Impact

- An estimated **70–90%** reduction in recurring manual preparation time, with no loss in accuracy
- Consistent, governed data ingestion every quarter
- Removed the main source of schema-related errors and rework
- Enabled automation-ready workflows for chargeback analysis and reporting
- Created a reusable ingestion pattern for future datasets